

Classification of EEG Signals by using Support Vector Machines

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Abstract—In this work, EEG signals were classified by support vector machines to detect whether a subject's planning to perform a task or not. Various different kernels were utilized to find the best kernel function and after that, a feature selection process was realized. The results are comparable to the recent works.

Keywords — EEG, support vector machines, feature selection.

I. INTRODUCTION

In recent years, it became possible to understand the brain signals [1]. By interfacing the brains signals directly, it is possible to design brain computer interfaces to control devices without mechanical interfaces. Especially direct controlling the prosthetic organs is very important for disabled people. The most common brain activity monitoring device is electroencephalogram (EEG).

In recent works, it was shown that mu rhythm which located in 7-13 Hz frequency band and central beta rhythm which located over 13 Hz band are originated from sensorimotor cortex and highly related to planning a movement [2].

Interpretation of brain activities by using electrical signals is a complex problem. Khare and friends [3] classified EEG recordings of four different mental states by using five classifiers. In this work, the best result was obtained by using a resilient back propagation method as 95%. Garrett and colleagues [4] were classified five different mental activities with support vector machines. In this work, maximum accuracy was realized as 72 per cent. Kerkeni and others [5] developed a system based on multilayer perceptron (MLP) to recognize a person's whether sleeping or awake by using EEG signals. Their system correctly classified 76 per cent of the EEG signals. Aljazeera and colleagues classified the EEG signals by using quantum neural network. In their paper they reported an accuracy of 81.33% [6]. Nandish et al [7] classified EEG signals with neural networks. Their system classified 80 per cent of the data correctly. Bhatt and Gopal [8] developed fuzzy classifiers to solve the EEG classification problem. In their work, they classified the Planning Relax data 71% correctly by using ID3, and 76.2% by using fuzz rough ID.

In this work the Planning Relax dataset located at University of California Irvine (UCI) Machine Learning Repository [9] is considered. Data set was created by using EEG signals collected from some volunteers. To find a better

solution, feature selection algorithms were applied to the dataset. Classification was done by using support vector machines (SVM). Section 2 describes the SVM. In Section 3, data set was introduced. In Section 4, feature selection algorithms were presented. Section 5 summarizes the results and conclusions.

II. SUPPORT VECTOR MACHINES

Support vector machine (SVM) is a statistical learning theory based classification method. For a given two-class linearly separable classification problem, SVM tries to find a hyperplane which separates the input space with a maximum margin. The optimum hyperplane is found as follows:

$$w \cdot x_i + b \geq +1, \text{ if } y_i = +1 \quad (1)$$

$$w \cdot x_i + b \leq -1, \text{ if } y_i = -1 \quad (2)$$

where x_i is the i^{th} input vector ($x \in R^N$), y_i is the class label of the i^{th} input ($y \in \{-1, +1\}$), w is the weight vector which normal to the hyperplane, and the b is the bias. Optimal hyperplane is found by two margins which parallel to the optimal hyperplane. Margins are found by Eq. 3 [10, 11]:

$$w \cdot x_i + b \leq \pm 1 \quad (3)$$

The input vectors that determine the margins are called as support vectors. If the problem is not linearly separable, a kernel function should be applied to the input vectors so the problem will become linearly separable in a transformed space:

$$K(x_i, x_j) = \varphi(x) \varphi(x_j) \quad (4)$$

Solution of a two-class linearly non-separable problem is as follows:

$$f(x) = \text{sign}(\sum_i \alpha_i y_i \varphi(x) \varphi(x_i) + b) \quad (5)$$

III. DATASET

In this work, Planning Relax dataset which is taken from UCI Machine Learning Repository [9] is utilized. The dataset consists of 12 measurements taken from EEG signals of a healthy subject. Five minutes long EEG records were taken while the subject was resting, and eyes were shut. Then, a beep sound of 60 dB was heard and the subject was asked to plan to move the right hand thumb for five seconds [9]. Eight EEG electrodes (C3, C4, P3, P4, F3, F4, T3, and T4) were placed according to the international standard 10-20 system [11] of electrode placement. Reference electrodes were placed on the

right and left ears, and the ground electrode was placed on the forehead. To remove the eye movement noise, two electrodes were placed on the outer canthus of the left and right eyes. EEG signals were filtered by a bandpass filter located in 1.6-50 Hz band. A notch filter was placed on 50 Hz to remove the artifacts caused by power line. All records were taken with a 70 mV/mm sensitivity and 256 Hz sampling frequency [9].

The collected EEG signals were decomposed to 6 levels, which have equal bandwidths, by using wavelet packet transform [13]. The signal was reconstructed by using 6th level coefficients to obtain the 7-13 Hz band [9].

IV. FEATURE SELECTION

Feature selection methods aim to remove irrelevant or less relevant features from the dataset to reduce the size of the data, and to improve the classification accuracy. In the literature, there are various feature selection algorithms [14, 15] which are collected into two groups. The first group which called as filters deals each features independently from the others. Due to their nature, filters are fast and computationally cheap. On the other hand, filter methods can't interpret the relations between features, and this disability limits the filters' performance. Wrappers consist of a searching algorithm and a classifier. The search algorithm searches new solutions through the feature space while the classifier produces a fitness function to the search algorithm. The most common algorithms are genetic algorithms, particle swarm optimization, etc. In this work, two filters and two wrappers were utilized.

Sequential forward selection (SFS) starts with an empty set. In every step, unused features are added to the existing solution set one by one and the one feature which improves the accuracy best is added to the solution. The algorithm stops when a predefined stopping criterion was reached. The stopping criterion should be maximum number of selected features, minimum accuracy, etc [14].

Sequential backward elimination is the opposite of SFS. The algorithm starts with the entire dataset. In every step, existing features are removed from the dataset one by one. The feature that gives the least decrease on the accuracy is excluded from the dataset. The algorithm continues until the stopping criterion is reached [14].

T-score is derived from t-test [15, 16]. For a given feature X, t-score is calculated as follows:

$$t(X) = \frac{|\mu_1 - \mu_2|}{\sqrt{((n_1\sigma_1^2 + n_2\sigma_2^2)/(n_1 + n_2))}} \quad (6)$$

where μ_i is the mean of X in the i^{th} class, σ_i is the standard deviation of X in the i^{th} class, n_i is the number of the instances in the i^{th} class respectively. Once the t-scores are calculated, N features which have greater t-scores, or the features have t-scores greater than a predefined threshold are selected. P-value is calculated as follows.

$$p(X) = \frac{|\mu_1 - \mu_2|}{\sigma_1 - \rho_2} \quad (7)$$

V. SIMULATION AND RESULTS

In this work, Planning Relax dataset taken from UCI Machine Learning Repository [2] is considered. The dataset consists of 182 samples. Each sample contains 12 features. The data divided into two equal parts as training and test data. All of the experiments realized ten times, and averaged to find the accuracies.

In the first step of the work, the dataset classified by linear and nonlinear SVMs, and the best kernel function was determined. Based on the results, the best choice is radial basis function (RBF) kernel for $\sigma=1.2$ (Table 1). Then, the dataset was reduced by using feature selection methods by using this configuration.

TABLE I. PERFORMANCE OF SVM WITH DIFFERENT KERNELS

	Linear	Quadratic	Polynomial	Radial Basis Function
Parameter	–	–	order : 5	$\sigma=1.2$
Accuracy	% 67.03	%58.24	%65.27	%71.43

By applying the SFS on data, the best result is obtained by only using the feature #12. With this feature, the accuracy was realized as 72.53 per cent (Table 2). After the SFS, SBE algorithm applied to the dataset. As seen in the Table 3, SBE gave the best result as 74.73% by using 11 features. In this simulation, only the feature #2 was omitted.

TABLE II. ACCURACIES OBTAINED BY SFS

# of features	Accuracy %	Selected features
1	72.53	12
2	71.79	4, 12
2	71.79	9, 12
3	71.43	7, 9, 12
3	71.43	9, 11, 12
4	71.43	1, 9, 11, 12

At the last stage of the simulations, t-scores and p-values of the features were calculated and the best features were selected according these metrics. Tables 4 and 5 summarize these results.

TABLE III. ACCURACIES OBTAINED BY SFS

# of features	Accuracy %	Rejected features
11	74.73	2
10	70.33	2, 5
9	73.63	2, 5, 12
8	72.54	2, 5, 11, 12

TABLE IV. ACCURACIES OBTAINED BY T-SCORE

# of features	Accuracy %	Selected features
1	71.43	4
2	71.43	4-10
3	68.13	4-10-9
4	67.03	4-10-9-5

TABLE V. ACCURACIES OBTAINED BY P-VALUE

# of features	Accuracy %	Selected features
1	71.43	4
2	69.23	4-9
3	69.23	4-9-7
4	70.33	4-9-7-5

VI. CONCLUSION

In this work, Planning Relax dataset taken from UCI Machine Learning Repository [2] is classified by using SVM. To raise the accuracy and to find the most relevant features, four different feature selection algorithms applied to the dataset. To find the most suitable kernel, different SVMs were trained with original data. The best accuracy for the original data was reached by RBF kernel as 71.43%. By applying SFS on the data, the accuracy reached to 72.53% with only one feature. The best result was obtained by SBE as 74.73% with eleven features, which is close to but worse than [8]. The best accuracies obtained by T-score and p-value are equal to the original data. The results show that the lower frequency band of 7-13 Hz bandwidth contains less information on movement planning. Therefore, in the future works, it is worthy to investigate the mid- and higher parts of the bandwidth.

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